

How the PaintOmics AI analysis is conducted

One round = 4 runs. 26 rounds so far, 57 runs kept. Everything below is in the repo: src/benchmarks/ai_arm_bench.py

A — INPUT: two jobs, built once, reused every round

STATegra multi-omics
examplefiles/datasets/08-stategra-multiomics
5 omic layers, 6 time points

ai_arm_bench jobs 2
POST /pa_step1 → /pa_step2
→ 731734364H, 1354co025T

Why the same two jobs every round
The pathway data is then identical across rounds, so a difference between arms is the arm and not the biology. The interpretation record is overwritten each run, which is why the metrics and the report are copied out.

B — THE ROUND: 4 runs, interleaved

gateway probe
refuses to start if the model service is
down

base-r1

agent-r1

base-r2

agent-r2

Interleaved on purpose. Gateway throughput drifts over tens of minutes. Running one arm's replicates back to back lets that weather land entirely on one side and look like a real effect.

Each run is in-process, ceiling 600 s. AI_FULL_AGENT=1 selects the agent arm; everything else is identical.

C — WHAT ONE RUN LEAVES BEHIND

metrics JSON
wall_s, citations, redactions,
coverage, papers, chars

report .md
the prose itself — metrics
cannot say if it reads well

trace JSONL (per run)
__config__ fingerprint · every tool call
with args/result/ms · __outcome__

MongoDB record
report, papers, verification, stats
(one per JOB — overwritten each run)

The trace is the only per-RUN record: Mongo keeps one interpretation per JOB, so with two jobs reused for 57 runs it can answer "how did this job's last run go" and nothing else.

D — HOW EACH NUMBER IS DERIVED

prose = report up to the FIRST appended table
("## Enriched Pathway Summary", "## Pathway Clusters", "### References")

citations = distinct [N] in the body that resolve to a retrieved paper

redactions = verification.redacted_count — sentences+entries the gate removed

coverage = indexed pathway names that appear in the PROSE

The trap this metric already fell into. The first version counted a pathway as covered if its name appeared anywhere in the report. Both arms APPEND a table of pathway names, so the agent arm scored 102/102 by printing a table it had never analysed. The prose cut is now permanent and has its own tests.

E — THE VERDICT: five rules, written before any arm ran

1 every replicate done, wall ≤ 600 s
2 citations ≥ base
3 redactions ≤ base + 2
4 prose coverage ≥ base
5 length within [0.6×, 2.0×] of base

ALL five must hold, or the incumbent stands.

Runs are grouped by **code fingerprint** — a hash of the module source, the Lead prompt and every tool description. Grouping changes the answer, so it is fixed in advance: agent-v20 alone fails rule 3, pooled with v19 it passes.

10 tests pin the rules and the prose cut, so they cannot drift once the numbers exist.

F — WHAT THE TRACES ARE FOR: diagnosis, not scoring

ai_tool_usage
adoption, cost per call,
outcome per tool

ai_citation_audit
integrity · quotes · pubmed
· structure

per-event timeline
overlap, gaps, who waited
on whom

→ a diagnosis → ONE change → new fingerprint → next round

Example, round 25 → 26: the score said "not better". The TRACE said why — r1 and r2 differed on one thing, whether the merged text's citations could be grounded in the time left (r1: 11 citations / 0 redactions; r2: 0 / 44). Scores rank arms; traces explain them. Only the trace tells you what to change next.

G — GUARDS AGAINST FOOLING MYSELF

Pre-register the prediction AND the falsifier
Round 26 was written down before it ran: redactions down, citations level or up — and "citations falling alongside redactions" named in advance as the result that would mean the diagnosis was wrong.

Never start against a broken instrument
Two replicates once burned ten minutes each against a gateway returning 504 and produced two outage reports dressed as results. The round now refuses to start, and the probe names WHICH layer failed.

Split stored data by era before pooling
The database holds several code eras at once. Three conclusions reversed when the same data was split by date instead of pooled — a "live" redaction defect that was March-only, missing telemetry that predated the stat, and a retrieval fix that the pooled average argued FOR and the split argued against.

A ratio is not a measurement
47 s of verifier work inside a 55 s window looked like 0.9× concurrency, and I drew it as a defect with a named cause. Computing OVERLAP from the per-event intervals showed two bursts of eleven. The span had contained something else: a 48 s rewrite between rounds.

Keep the raw runs, not just the conclusions
57 per-run metric files live in the repo. The scorer reproduces every verdict in the write-up, including the ones that went against the arm I was building.

One change per round
Nine changes accumulated during a gateway outage and had to be measured as a bundle, which is why round 25 could rank the arm but not attribute the result. The fingerprint at least says WHICH agent was measured.

n=2 is a story, not a measurement
Two replicates of identical code once came out 47 094 and 10 043 characters of prose, and 5 vs 10 redactions. A two-run mean is reported as a two-run mean; the replicate spread is always shown beside it.

Verify the fix RAN before believing the result
Round 26's grounding step never executed: the guard tested for [N] markers, but sub-agents emit (PMID: X) until the merge converts them. The trace showed no delegate_grounding event at all — which is how it was caught, rather than being read as "the fix did not help".